MATH 112B

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Spring 2026

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<u>Conservation Laws</u>
This is used to describe: bacterial movement in a petridish, nerve signal traveling through an axon, heat diffusion in a rod, cars on the highway.

We start with the following assumptions:

- Motion happens in 1D.
- Cross-sectional area is constant.

Consider a 1D tube, with particles inside. How do we describe the change of concentration of particles?

$$\begin{matrix}\text{change in pop between} \\ x \text{ and } x+\Delta x\end{matrix} = \begin{matrix}\text{rate of entry into} \\ (x, x+\Delta x) \\ \text{per unit time}\end{matrix} - \begin{matrix}\text{rate of exit into} \\ (x, x+\Delta x) \\ \text{per unit time}\end{matrix}$$

Define

- $c(x,t)$ = concentration of particles at x at time t
- $J(x,t)$ = flux of particles at (x,t) = # of particles crossing a unit area at x in the positive direction, per unit time.
- A = cross-sectional area of the tube.
- ΔV = volume of our segment = $A \cdot \Delta x$

We have

$$\begin{aligned}\frac{\partial}{\partial t}(\underbrace{c(x,t)\cdot A\cdot \Delta x}_{\#\text{ of particles}}) &= J(x,t)A - J(x+\Delta x,t)A \\ \frac{\partial c}{\partial t} &= \frac{J(x,t)-J(x+\Delta x,t)}{\Delta x} \\ \lim_{\Delta x\to 0}\frac{\partial c}{\partial t} &= \lim_{\Delta x\to 0}\frac{J(x,t)-J(x+\Delta x,t)}{\Delta x} \\ \frac{\partial c}{\partial t} &= -\frac{\partial J}{\partial x} \;\leftarrow \text{Conservation of mass}\end{aligned}$$

Consider a new term,

$$\begin{matrix}\Delta\text{pop between} \\ x \text{ and } x+\Delta x\end{matrix} = \begin{matrix}\text{rate of entry into} \\ (x, x+\Delta x) \\ \text{per unit time}\end{matrix} - \begin{matrix}\text{rate of exit into} \\ (x, x+\Delta x) \\ \text{per unit time}\end{matrix} \pm \begin{matrix}\text{rate of local} \\ \text{creation} \\ \text{or degradation}\end{matrix}$$

We now have

- $\sigma(x,t)$ = # of particles created/removed per unit volume at time t

Now our conservation laws become

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x} \pm \sigma(x,t)$$

To generalize to 2D or 3D:

$$\frac{\partial c}{\partial t} = -\vec{\nabla}\vec{J} \pm \sigma(\vec{x}, t)$$
$$\text{Now we have } c = c(\vec{x}, t)$$
$$\sigma = \sigma(\vec{x}, t)$$
$$\vec{J} = (J_x, J_y, J_z)$$
$$\vec{\nabla} = (\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z})$$

So we have

$$\frac{\partial c}{\partial t} = -(\frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z}) \pm \sigma(x, y, z, t)$$

$\boxed{\text{ex}}$ Consider a moving fluid + organisms that move with the flow (not actively). Suppose that the velocity of the fluid is given by

$$\vec{v}(x, y, z, t)$$

Let $c(x, y, z, t)$ = concentration of organism.

What is the flux?

$$\vec{J} = c \cdot \vec{V} \quad \text{Assume } \sigma = 0$$
$$\Rightarrow \frac{\partial c}{\partial t} = -\left(\frac{\partial cV_x}{\partial x} + \frac{\partial cV_y}{\partial y} + \frac{\partial cV_z}{\partial z}\right)$$

In 1D:

$$\begin{aligned}\frac{\partial c}{\partial t} &= -\frac{\partial}{\partial x}(cV) \\ &= -\frac{\partial c}{\partial x}V - \frac{\partial V}{\partial x}c\end{aligned}$$

What kind of PDE is this?

- 1st order
- 2 variables: x, t
- Linear

- Transport equation

$\boxed{\text{ex}}$ Attraction/repulsion. Suppose that ψ is a source of attraction for the bacteria. The motion is determined by the gradient of ψ.
In 1D: $\psi = \psi(x,t),\ J = \frac{\partial \psi}{\partial x}$

$$J = c\alpha \vec{\nabla}\psi$$

We have

$$\begin{aligned} \frac{\partial c}{\partial t} &= -\frac{\partial J}{\partial X} \\ \frac{\partial c}{\partial t} &= -\frac{\partial}{\partial t}\left(c\alpha\frac{\partial \psi}{\psi x}\right) \end{aligned}$$

ψ is given.

$$\frac{\partial c}{\partial t} = -\alpha\frac{\partial c}{\partial x}\cdot\frac{\partial \psi}{\partial x} - \alpha c\frac{\partial c}{\partial x}\cdot\frac{\partial^2 \psi}{\partial x^2}$$

This is 1st order, linear PDE in 2 variables.

$\boxed{\text{ex}}$ Temperature of a heated rod.
Here we define $c(x,t)$ as concentration of energy/temperature. And since heat spreads from hot to cold, we get the following.

$$\boxed{\text{Fick's Law} : J = -D\cdot\frac{\partial c}{\partial x}}$$

Then we have

$$\begin{aligned} \frac{\partial c}{\partial t} &= -\frac{\partial}{\partial x}\left(-D\frac{\partial c}{\partial x}\right) \\ &= D\frac{\partial^2 c}{\partial x^2} \ \leftarrow \text{Heat equation} \end{aligned}$$

In higher dimensions:

$$\begin{aligned} \frac{\partial c}{\partial t} &= \vec{\nabla}(D\vec{\nabla}c) \\ &= D\left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} + \frac{\partial^2 c}{\partial z^2}\right) \end{aligned}$$

$\boxed{\text{ex}}$ Random motion of particles.
Denote λ_r as the amount of particles jumping from $x \to x + \Delta x$
Denote λ_l as the amount of particles jumping from $x \to x - \Delta x$
Denote $c(x,t) = \#$ of particles within $(x, x+\Delta x)$ at time t:

$$c(x,t+\Delta t) = c(x,t) + \lambda_r \cdot c(x-\Delta x,t) + \lambda_l \cdot c(x+\Delta x,t) - \lambda_l c(x,t) - \lambda_r c(x,t)$$

Using Taylor Series:

$$\begin{aligned}
c(x,t+\Delta t) &= c(x,t) + \frac{\partial c}{\partial t}\Delta t + \frac{1}{2}\frac{\partial^2 c}{\partial t^2}(\Delta t)^2 + \dots \\
c(x+\Delta x,t) &= c(x,t) + \frac{\partial c}{\partial x}\Delta x + \frac{1}{2}\frac{\partial^2 c}{\partial x^2}(\Delta x)^2 + \dots \\
c(x-\Delta x,t) &= c(x,t) - \frac{\partial c}{\partial x}\Delta x + \frac{1}{2}\frac{\partial^2 c}{\partial x^2}(\Delta x)^2 + \dots
\end{aligned}$$

We can simplify using $\lambda_r = \lambda_l = \lambda$, then we have

$$\begin{aligned}
c + \frac{\partial c}{\partial t}\Delta t + \frac{\partial^2 c}{\partial t^2}\frac{(\Delta t)^2}{2} + \dots = c & \\
&+ \lambda\left(c + \frac{\partial c}{\partial x}\Delta x + \frac{1}{2}\frac{\partial^2 c}{\partial x^2}(\Delta x)^2\right) \\
&+ \lambda\left(c - \frac{\partial c}{\partial x}\Delta x + \frac{1}{2}\frac{\partial^2 c}{\partial x^2}(\Delta x)^2\right) \\
&- \lambda c - \lambda c \\
\frac{\partial c}{\partial t}\Delta t + \frac{\partial^2 c}{\partial t^2}\frac{(\Delta t)^2}{2} &= \lambda\frac{\partial^2 c}{\partial x^2}(\Delta x)^2
\end{aligned}$$

Call $\lim_{\Delta x \to 0, \Delta t \to 0} \frac{(\Delta x)^2}{\Delta t} \equiv K$ and ignore second order time term as we take the limit.

$$\begin{aligned}
\frac{\partial c}{\partial t}\Delta t &= \lambda(\Delta x)^2 \cdot \frac{\partial^2 c}{\partial x^2} \\
\frac{\partial c}{\partial t} &= \lambda K \cdot \frac{\partial^2 c}{\partial x^2} \ \leftarrow \text{Diffusion equation}
\end{aligned}$$